

Practical No : 6

Tutorial Title: Write a program in C++ to perform the following operations on complex numbers: Add, Subtract, Multiply and Divide. Use operator overloading for these operations.

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Program:

```
#include <iostream>
using namespace std;
class Complex {
    float real, imag;
public:
    void read() {
        cin >> real >> imag; }
    void display() {
        if (imag >= 0)
            cout << real << " + " << imag << "i";
        else
            cout << real << " - " << -imag << "i"; }
    Complex operator+(Complex c) {
        Complex t;
        t.real = real + c.real;
        t.imag = imag + c.imag;
        return t; }
    Complex operator-(Complex c) {
        Complex t;
        t.real = real - c.real;
        t.imag = imag - c.imag;
        return t; }
    Complex operator*(Complex c) {
        Complex t;
        t.real = real * c.real - imag * c.imag;
        t.imag = real * c.imag + imag * c.real;
        return t; }
    Complex operator/(Complex c) {
        Complex t;
```

```

        float d = c.real * c.real + c.imag * c.imag;
        t.real = (real * c.real + imag * c.imag) / d;
        t.imag = (imag * c.real - real * c.imag) / d;
        return t; }
};

int main() {
    Complex c1, c2;
    Complex add, sub, mul, div;
    cout << "Enter first complex number (real imaginary): ";
    c1.read();
    cout << "Enter second complex number (real imaginary): ";
    c2.read();
    add = c1 + c2;
    sub = c1 - c2;
    mul = c1 * c2;
    div = c1 / c2;
    cout << "\nAddition = ";
    add.display();
    cout << "\nSubtraction = ";
    sub.display();
    cout << "\nMultiplication = ";
    mul.display();
    cout << "\nDivision = ";
    div.display();
    return 0;
}

```

Output:

```
main.cpp
1 #include <iostream>
2 using namespace std;
3 class Complex {
4     float real, imag;
5 public:
6     void read() {
7         cin >> real >> imag;
8     }
9     void display() {
10         if (imag >= 0)
11             cout << real << " + " << imag << "i";
12         else
13             cout << real << " - " << -imag << "i";
14     }
15     Complex operator+(Complex c) {
16         Complex t;
17         t.real = real + c.real;
18         t.imag = imag + c.imag;
19         return t;
20     }
21     Complex operator-(Complex c) {
22         Complex t;
23         t.real = real - c.real;
24         t.imag = imag - c.imag;
25         return t;
26     }
27     Complex operator*(Complex c) {
28         Complex t;
29         t.real = real * c.real - imag * c.imag;
30         t.imag = real * c.imag + imag * c.real;
31         return t;
32     }
33     Complex operator/(Complex c) {
34         Complex t;
35         float d = c.real * c.real + c.imag * c.imag;
36         t.real = (real * c.real - imag * c.imag) / d;
37         t.imag = (imag * c.real - real * c.imag) / d;
38         return t;
39     }
40 };
41 int main() {
42     Complex c1, c2;
43     Complex add, sub, mul, div;
44     cout << "Enter first complex number (real imaginary): ";
45     c1.read();
46     cout << "Enter second complex number (real imaginary): ";
47     c2.read();
48     add = c1 + c2;
49     sub = c1 - c2;
50     mul = c1 * c2;
51     div = c1 / c2;
52     cout << "\nAddition = ";
53     add.display();
54     cout << "\nSubtraction = ";
55     sub.display();
56     cout << "\nMultiplication = ";
57     mul.display();
58     cout << "\nDivision = ";
59     div.display();
60     return 0;
61 }
```

Output

```
Enter first complex number (real imaginary): 4
5
Enter second complex number (real imaginary): 2
5

Addition = 6 + 10i
Subtraction = 2 + 0i
Multiplication = -17 + 30i
Division = 1.13793 - 0.344828i

*** Code Execution Successful ***
```